

Jet-noise line-sources extraction from Particle Image Velocimetry data using Hilbert Proper Orthogonal Decomposition

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In this work, a technique for identifying wavepacket sound sources in turbulent subsonic jets from velocimetry data is proposed. The technique is based on a space-only variant of the Hilbert Proper Orthogonal Decomposition, allowing the extraction of wave-like patterns from temporally under-resolved data by leveraging the equivalence between time and convective direction for traveling waves. The velocity field patterns retrieved are then employed in a Galerkin projection of Lighthill’s equation source term to determine their contribution to the sound source. The sound sources identified are finally reduced to a line-source wavepacket model and used to estimate their contribution to the emission pattern of the jet noise. The method is tested on numerical data from a Large Eddy Simulation of a jet at $M = 0.9$. The availability of time-resolved data from this dataset allows the identification of the estimated wavepackets’ behavior in time, proving that they are consistent with advecting flow structures. Differently from other techniques such as Spectral Proper Orthogonal Decomposition, the structures are not time-periodic. Instead, they are characterized by modulation of their frequency through time, i.e. they are coherent over a short period and time-intermittent. In this sense, the technique represents a time-frequency analysis of the flow structures (akin to a short-time Fourier transform) rather than a pure spectral one (akin to a Fourier transform). The temporal information also allows the extraction of characteristic frequencies and convective velocities of the source contributions, providing full access to the estimated emission patterns. The methodology can also be applied to non-time-resolved data, still providing access to the spatial information of the source contribution. While this information is not enough to fully estimate the noise emission, it can still provide a measure of the potential emission, and thus of the relevance from an acoustic viewpoint, of the source contribution.

I. Introduction

Despite being a focal point of extensive research in aeroacoustics, the mechanisms underlying jet noise still defy complete comprehension within the scientific community. While the work by Lighthill identified chaotic turbulent eddies in the subsonic jet shear layer as the primary source of mixing noise [1], consensus in the literature points to the crucial role played by coherent structures, commonly referred to as wave-packets, which advect with the flow and contribute significantly to sound radiation [2]. Despite constituting a small fraction of the total turbulent kinetic energy [3, 4], these structures exhibit enhanced sound emission compared to smaller turbulence scales, making them a notable contributor to noise despite their lower energy content. Crighton and Huerre [5] demonstrated that the sound emitted by subsonically advecting coherent structures is intricately linked to their non-homogeneity in the advection direction, particularly their modulation in the axial direction of the jet, resulting in high acoustic efficiency at low zenithal angles. It is widely accepted – albeit without conclusive proof – that shock-free jet noise arises from two distinct sources: an omnidirectional incoherent source associated with fine-scale turbulence and a coherent source linked to wave-packets predominantly radiating at small zenithal angles [6].

Lighthill’s acoustic analogy [1] has been used to explain jet noise starting from near-field quantities. This analogy is

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formulated as

$$\frac{\partial^2 \rho}{\partial t^2} - c_0^2 \nabla \rho = \frac{\partial^2 T_{ij}}{\partial x_i \partial x_j} = S_{i,j}, \quad (1)$$

where $T_{ij} = \rho u_i u_j + \delta_{ij}(p - c_0^2 \rho) + \tau_{ij}$ is the Lighthill's stress tensor, p is the pressure, ρ is the fluid density, c_0 is the speed of sound, τ_{ij} are the viscous stresses, u_i are the velocity components in the spatial direction x_i and t is time. Einstein's summation is implied in the previous formula. The right-hand side of Eq. (1) $S_{i,j}$ is the sound source term for the non-homogeneous wave equation on the left-hand side and depends on the near-field characteristics of the flow. The term $\rho u_i u_j$ dominates the sound source for unheated jets [7].

Based on the experimental observations of jet near-field measurements, several source models were proposed for non-compact wave-packet sources [8]. For example, Crow [9] proposed the line-source model

$$S_{xx}(y) = \exp(-i\omega t) \exp(ik_h y) \exp\left(\frac{-y^2}{L^2}\right), \quad (2)$$

where ω is the temporal frequency, k_h is the hydrodynamic wavenumber and L the length scale of the wavepacket amplitude envelope. The acoustic pressure power spectral density for a listener located at a distance D from the source and at an angle θ from the jet axis is given by

$$\langle p(D, \omega) p^*(D, \omega) \rangle \approx \frac{1}{16\pi |D - y|^2} L^2 \exp\left(-\frac{1}{2} k_h^2 L^2 (1 - M_c \cos \theta)^2\right) \quad (3)$$

with M_c being the convective Mach number [8].

The aeroacoustic community currently dedicates significant efforts to identify elusive wave-packets [10] and develop models for their sound emission [8]. Identifying wave-packet structures enables a *divide and conquer* strategy in tackling the noise generation problem, allowing for an assessment of their contribution to overall noise emission. The primary challenges arise from the low energetic content of these features and their temporal intermittency, making it difficult to isolate them non-intrusively, i.e. without external forcing on the jet [11]. Recent attempts to identify the distribution of wave-packets in unforced turbulent jets employ advanced post-processing techniques. Spectral Proper Orthogonal Decomposition (SPOD, [12]) stands out as a leading technique, isolating complex-valued wave-packet shapes in flow fields by leveraging their spatio-temporal coherence achieved through Fourier decomposition of time series. While SPOD provides a spectrally pure decomposition, features resembling traveling waves can be identified as pairs of real-valued modes sharing a phase quadrature relationship in both time and space. These can be easily detected by standard Proper Orthogonal Decomposition (POD, [13]) when they contribute significantly to turbulent energy, although the phase quadrature relationship may be compromised for lower energy structures. Addressing this, Kriegseis *et al.* [14] introduced Hilbert POD (HPOD), an extension of POD that automatically yields complex-valued modes, both in time and space, whose real and imaginary components satisfy a phase quadrature relationship in time due to the properties of the Hilbert transform. However, a shared drawback with SPOD is the necessity for time-resolved series to compute the Hilbert transform (or Fourier transform in the case of SPOD). While this is commonly achieved by expensive high-fidelity Large Eddy Simulations (LES) and time-resolved Particle Image Velocimetry (PIV) experiments, these techniques cannot be applied to data from snapshot PIV, which is more commonly available.

To overcome the above-mentioned shortcomings from limited temporal resolution, in the present work the use of a space-only variant of the HPOD to identify wave-packets in turbulent jets from velocity field data is proposed. This method has already been proven effective in identifying advecting wave-like structures in velocity fields [15]. The modes identified through HPOD are then utilized in a Galerkin projection of the source term provided by Lighthill's aeroacoustic analogy to identify equivalent sources. In the present analysis, these source contributions are simplified to line sources, modeled after Eq. (2), to assess their contribution to noise emission. The data used to test the method in the present work are time-resolved data from a turbulent jet Large Eddy Simulation at $M = 0.9$. The availability of a time-resolved sequence from this dataset allows to understand the temporal behavior of the extracted source modes, helping to confirm that they correspond to advecting features in the short-time frame. Additionally, access to temporal information allows the extraction of characteristic frequencies and advection velocities of the source contributions, which can be used for estimating the full noise emission. While this information would not be available from typical snapshot PIV experiments, the technique can still be applied to these data to extract spatial information, which in turn can provide useful insights into their potential acoustic relevance. Additionally, temporal information can be retrieved by pairing PIV with high-repetition-rate sensors (either pressure sensors or microphones, for example) using Extended-POD-based [16] (or more advanced [17]) methodologies for dynamic estimation of flow fields.

II. Mathematical background

A. Space-only Hilbert POD

The Hilbert POD is an extension of the standard POD, which allows the extraction of complex-valued modes from a real-valued dataset. This operation requires the *complexification* of the dataset before applying POD. This operation, for example, is carried out with Fourier transform for Spectral POD. In Hilbert POD this operation is performed, instead, by computing the analytic representation of the velocity field $\mathbf{u} + i\mathcal{H}[\mathbf{u}]$, with i being the imaginary unit and where the operator $\mathcal{H}$ indicates the Hilbert transform. While this operation is typically performed in the temporal direction, in the space-only HPOD the Hilbert transform is performed in the advection direction of the jet. Therefore, given the real-valued velocity field $\mathbf{u}$, the Hilbert transform $\mathcal{H}$ applied along the x dimension, assuming that this is the direction in which flow structures are advected, is given by

$$\mathcal{H}[\mathbf{u}(x, r, t)] = \lim_{e \rightarrow 0} \frac{1}{\pi} \int_{|\xi - x| > e} \frac{u(\xi, r, t)}{x - \xi} d\xi. \quad (4)$$

The Hilbert transform can be interpreted in the Fourier space as a $\pi/2$ shift, either in advance or delay depending on the sign of the frequency component. In the present work, the discrete Hilbert transform is computed by computing the Fast Fourier Transform along the x direction, setting to zero all the negative frequency terms and then applying the Inverse Fourier Transform. The snapshot POD method [18] is applied to compute the decomposition of the complex-valued analytic representation of $\mathbf{u}$ as

$$\mathbf{u}(x, r, t) + i\mathcal{H}[\mathbf{u}(x, r, t)] = \sum_m \psi^{(m)}(t) \sigma^{(m)} \phi^{(m)}(x, r), \quad (5)$$

where $\psi^{(m)}$ is the complex-valued HPOD temporal m -th mode, $\phi^{(m)}$ is the complex-valued HPOD spatial m -th mode and $\sigma^{(m)}$ is the real-valued m -th singular value. The objective of the space-only HPOD is to identify modes with a characteristic wave-packet shape

$$\psi^{(m)}(t) \phi^{(m)}(x, r) = \Psi^{(m)}(t) \Phi^{(m)}(x, r) e^{i(\lambda_m x - \omega_m t)}, \quad (6)$$

where $\Psi^{(m)}$ and $\Phi^{(m)}$ are low-frequency temporal and spatial envelopes modulating the wave. This is possible thanks to the properties of the analytic signal representation, where the imaginary part appended to the real-valued field is shifted by $\pi/2$ for all frequencies in the direction in which the Hilbert transform is applied. According to the wavepacket nature of the modes in Eq.(6), the shift can be either interpreted as a temporal shift or, equivalently, as a spatial shift in the advection direction:

$$e^{i(\lambda_m x - \omega_m t)} = f(x, t) + i f\left(x, t + \frac{1}{\omega_m} \frac{\pi}{2}\right) = f(x, t) + i f\left(x + \frac{1}{\lambda_m} \frac{\pi}{2}, t\right) \quad (7)$$

B. Galerkin projection of the sound source

The source term of the Lighthill's analogy for an unheated jet can be written as:

$$S_{i,j} = \frac{\partial^2 T_{ij}}{\partial x_i \partial x_j} = \rho \frac{\partial^2 u_i u_j}{\partial x_i \partial x_j} \quad (8)$$

Using the HPOD in Eq. (5), a Galerkin projection of the source term into the space of the HPOD modes is performed via

$$S_{i,j} = \sum_m \sum_n \sigma^{(m)} \sigma^{(n)} \psi^{(m)} \psi^{(n)} \phi_{S,m,n} = \rho \sum_m \sum_n \sigma^{(m)} \sigma^{(n)} \psi^{(m)} \psi^{(n)} \frac{\partial^2 \phi_i^{(m)} \phi_j^{(n)}}{\partial x_i \partial x_j}, \quad (9)$$

where $\phi_{S,m,n}$ represents the quadratic contribution to the source of the m -th and n -th modes and $\psi^{(m)} \psi^{(n)} = \psi_{S,m,n}$ represents the temporal evolution of the source contribution.

C. Line-source fitting

In order to identify line sources, the quadratic source contributions $\phi_{S,m,n}(x, r)$ have been integrated along the radial direction. The resulting functions $\int \phi_{S,m,n}(x, r) dr$ have been fitted according to the model in Eq. (2) in order to extract the hydrodynamic wavenumber k_h and the length scale of the wavepacket amplitude envelope L . Given the non-spectrally-pure nature of the source wavepackets, the hydrodynamic wavenumber has been approximated with the most energetic wavenumber in its spatial Fourier transform. The wavepacket lengthscale, instead, has been identified by fitting the wavepacket envelope $|\int \phi_{S,m,n}(x, r) dr|$ with a 5 terms Gauss model. The values obtained for each source contribution have been finally used to compute the estimated radiated noise according to Eq. (3).

III. Results

The line-source extraction method detailed above has been applied to a velocity field database, which has been extracted from the high-fidelity LES of a turbulent jet at $M_{jet} = 0.9$ and $Re_{jet} = U_{jet}d/\nu \approx 10^6$ provided in [19]. The database is constituted by 10000 time-resolved snapshots sampled at $\Delta t = 0.2d/c_0$. The data are available over a structured cylindrical grid spanning $30d$ in the axial direction and $6d$ in the radial one, containing (656,138,128) points in the axial, radial and azimuthal directions. For the application of the space-only HPOD, the axisymmetric component has been extracted for each snapshot. Space-only HPOD has been computed by performing the Hilbert transform along the axial direction x . In order to remove end-effects introduced by the Fourier transform, 15% of the flow field has been removed along x from both ends of the domain.

A. Space-only HPOD

The first 3 complex-valued modes are reported in Fig. 1. Given the complex nature of the modes, the spatial modes have been represented with their real (1st row), imaginary (2nd row) and absolute (3rd row) values. The modes show a clear wavepacket pattern, with real and imaginary part being shifted of $\pi/2$ in the streamwise direction. The absolute value represents the envelope of the wave with clearly defined multiple peaks at different spatial positions. A similar modulation pattern is also observable on the temporal SPOD modes recovered in [12] for a turbulent jet at $M = 0.4$.

To emphasize more clearly the wavepacket structure of the modes, in the first row of Fig. 2 the same 3 modes are reported after integration along the radial direction. In this figure the black continuous line represents the real part, the black dashed line the imaginary part and the red continuous line the wave envelope. In the second row, a portion of the temporal mode centered at an arbitrary instant is reported. Both the spatial and temporal modes are characterized by a wave-like pattern modulated by the wave envelope, clearly suggesting that the modes represent advecting wavepackets with a formulation as the one in Eq. (6). It must be highlighted that for the present test a time-resolved sequence has been used, from which the wave-like pattern of the temporal modes is revealed. However, the methodology applied does not require time-resolved data and produces similar spatial modes from snapshot data, as already proved by [15].

The temporal modes in Fig. 2 present a non-spectrally-pure behavior, being characterized by modulation and frequency variation on long periods of time. In order to analyze the temporal modes, time-frequency analysis is used. The spectrograms of the temporal modes are reported in the first row of Fig. 3 in terms of the Strouhal number $St = \frac{1}{2\pi} \frac{\omega d}{U_\infty}$, where ω is the temporal angular frequency. The spectrograms show that for each mode the spectral content varies over time but is localized in terms of Strouhal numbers. Also, the spectrogram shows that the modes are characterized by temporal intermittency, which reveals itself as periods in which there is no relevant spectral content. To characterize the frequency content through time, a histogram of the Power Spectral Density (PSD) peak frequencies is reported in the second row. For each mode, most of the peak frequencies are distributed around a specific frequency. Therefore, the frequency with the highest number of occurrences can be assumed as a reasonable estimate of the characteristic temporal frequency of each mode.

B. Line-source model

The space-only HPOD modes extracted in section III.A are used to estimate the contribution to the sound source following what is outlined in sections II.B and II.C. Fig. 4 shows a 3×3 matrix including the source contributions of 3 arbitrary HPOD modes (specifically the 2nd, 7th and 20th modes) computed from the Galerkin projection of the Lighthill's equation source term given by Eq. (9), each linked to a pair of velocity HPOD modes. For a clearer representation, the sources have been represented as line sources after integrating in the radial direction, which highlights their wavepacket nature. The line sources reported here show a wide variety of different behaviors in terms of shape, wavenumber, and lengthscales.

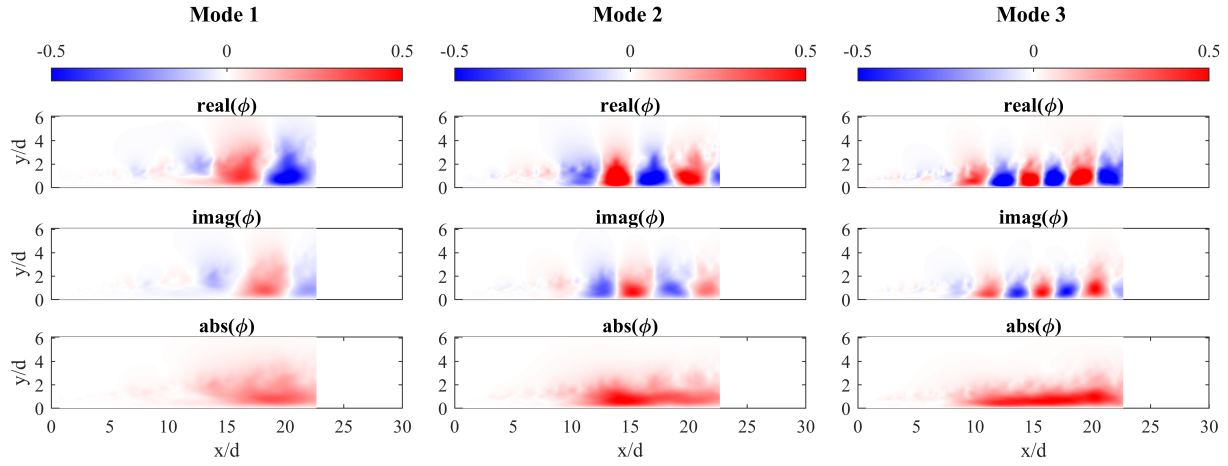


Fig. 1 Streamwise component of the first 3 complex-valued spatial modes computed from space-only HPOD: real part (1st row), imaginary part (2nd row), absolute value (3rd row).

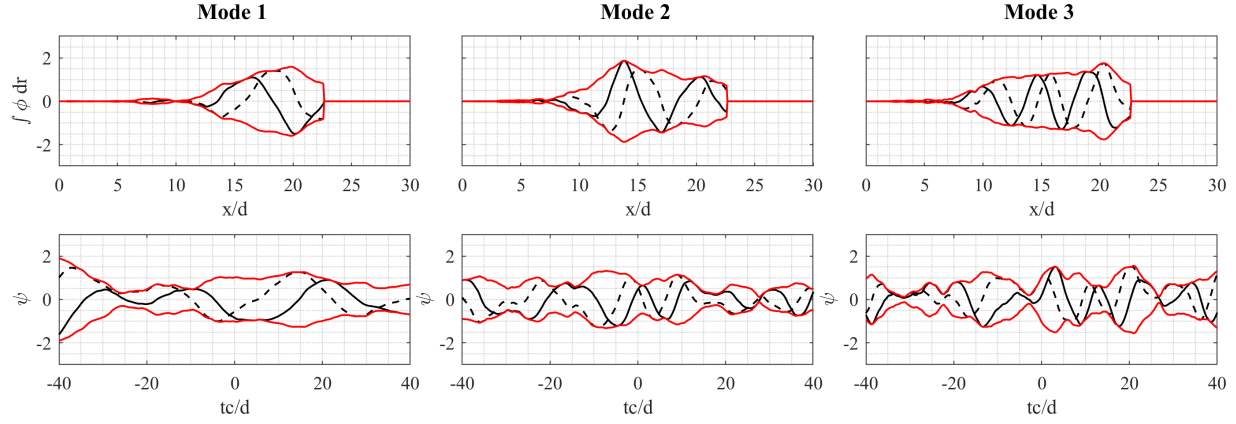


Fig. 2 First 3 complex-valued modes computed from space-only HPOD: radially-integrated spatial modes (top row); temporal modes (bottom row). The black continuous line represents the real part, the black dashed line the imaginary part and the red continuous line the wave envelope.

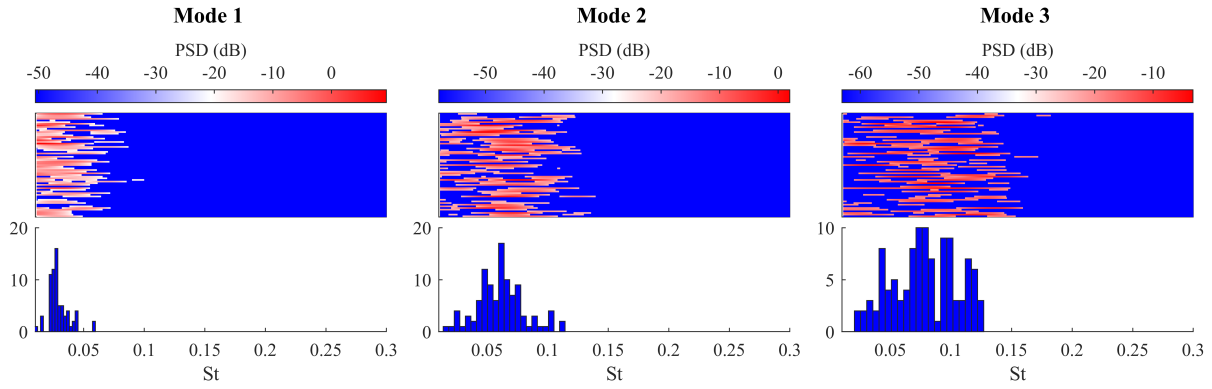


Fig. 3 First 3 complex-valued temporal modes computed from space-only HPOD: spectrogram (1st row), histogram of the peak PSD frequencies (2nd row).

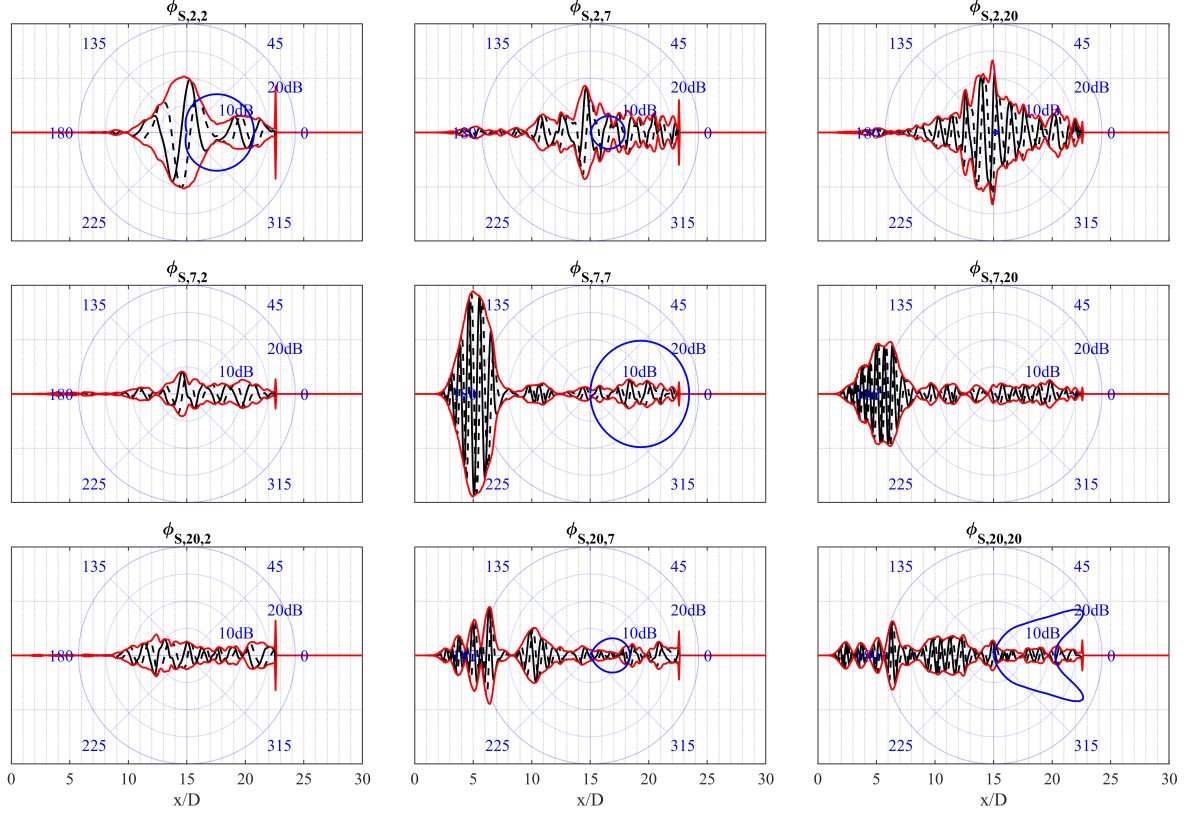


Fig. 4 Line-source contributions of the combination of 3 arbitrary HPOD modes computed from the Galerkin projection. The black continuous line represents the real part, the black dashed line the imaginary part and the red continuous line the wave envelope. The blue line, referring to the polar plot, represents the estimated SPL radiation pattern.

Each line-source contribution is overlaid with a polar plot representing their noise emission (in dB SPL) estimated as described in section II.C. For the emission estimation, a listener distance of $D = 50d$ has been used and the variation in the source position has been neglected. The representative convective Mach number has been computed from the wave phase velocity as $M_c = \frac{\omega_h}{k_h U_\infty}$, where ω_h is the representative angular frequency of the source-contribution temporal evolution $\psi_{S,i,j}$. This representative frequency has been extracted from the histogram of the peak PSD frequencies reported in Fig. 5.

The sound-emission patterns that emerge from the estimated wavepacket parameters show strongly different behaviors. The intensity of the emission is not directly linked to the energy content of the mode pair used to estimate the source contribution: relevant modes in terms of sound emission might have a generally low kinetic energy content, following a behavior that is well assessed in the literature about the topic [20, 21]. For example, the contribution $\phi_{S,7,2}$ has a much smaller sound emission than contribution $\phi_{S,20,20}$ – even though the modes 2nd and 7th have a higher kinetic energy content than mode 20th (notice that the HPOD modes, similarly to POD modes, are ranked by energy). This behavior is mostly dictated by the lengthscale of the wavepacket according to the model of Eq. 3. Also, depending on the spatial characteristics of the wavepacket and on the convective Mach number, vastly different emission patterns can be present. Contribution $\phi_{S,2,2}$, for instance, featuring a convective Mach number of about 0.42, presents a directivity pattern that enhances the emission at a 0 zenithal angle. Differently, contribution $\phi_{S,2,2}$, featuring a convective Mach number of about 1.11, enhances emission at an angle $\cos^{-1}(1/M_c)$ according to the model in Eq. 3. It is worth highlighting here that the source contributions proposed in Fig.4 do not correspond with the estimated loudest events, and serve only to provide a picture of the variability, which is displayed in the present analysis.

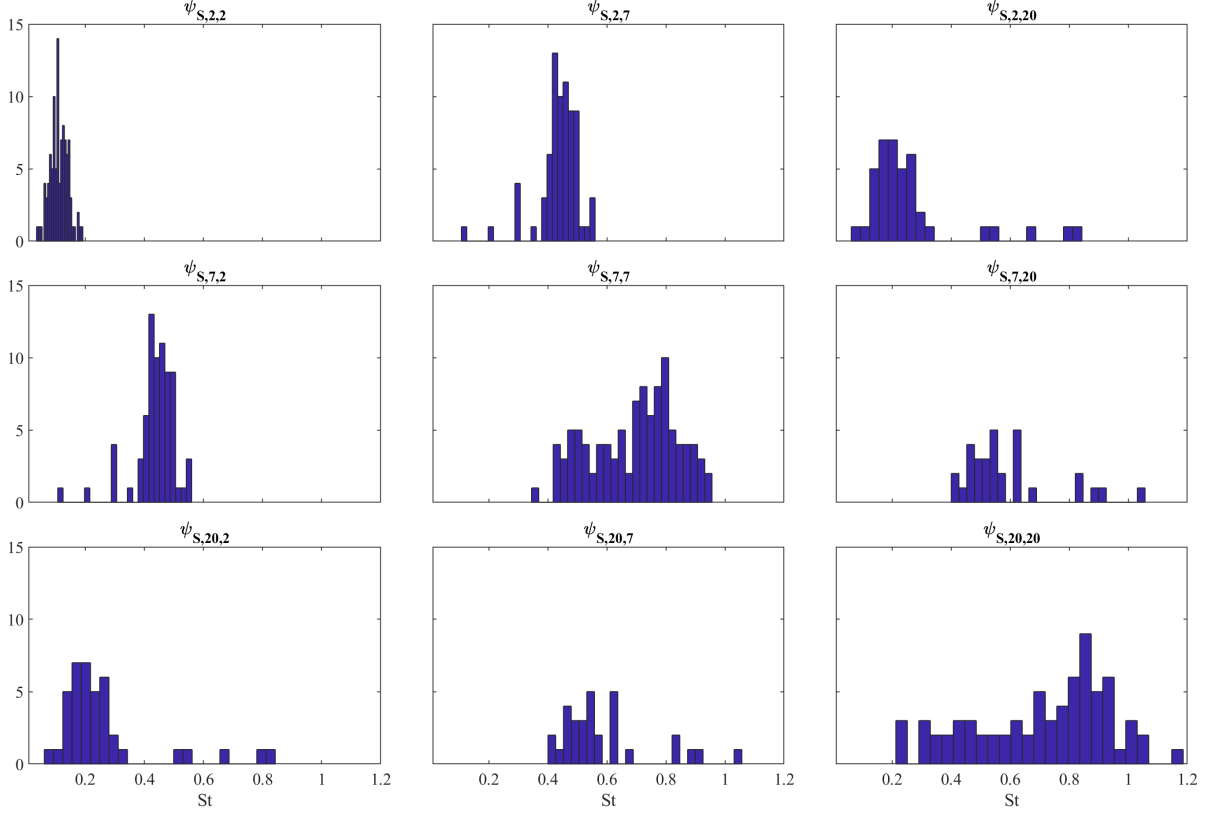


Fig. 5 Histogram of the peak PSD frequencies from the spectrogram of the combination of 3 arbitrary HPOD modes computed from the Galerkin projection.

IV. Conclusions

This work presents a novel methodology for identifying wavepacket sound sources in turbulent jets using space-only Hilbert Proper Orthogonal Decomposition (HPOD) and Galerkin projection of Lighthill's equation source term. This methodology can be readily applied to non-time-resolved datasets, which are typically more accessible in experimental setups.

The analysis as elaborated in the present work is performed on time-resolved data from an LES of a turbulent jet at $M = 0.9$ [19]. This allows the extraction of the temporal behavior of the space-only HPOD modes and confirms the wavepacket nature of such modes. The computed HPOD modes exhibit similarities to results obtained in analogous studies using Spectral Proper Orthogonal Decomposition (SPOD), indicating that space-only HPOD can be equally effective in extracting wave-like patterns from data. Differently from these techniques, however, the modes identified are not spectrally pure in time: the spectral content varies in time, as observed from time-frequency analysis, even if the peak frequency for each mode seems to statistically distribute around a specific characteristic value. Additionally, the time-frequency analysis reveals that the modes are characterized by periods of inactivity – i.e. literal "silence" in the present context – in which the spectral content goes to zero, and which can be interpreted as intermittency of the phenomenon. The sound source's contributions identified from the Galerkin projection of the Lighthill's equation have been reduced to a line-source model according to Eq. 2. The characteristics of the sound source contributions exhibit a broad range of variability in terms of lengthscale of wavepacket amplitude, hydrodynamic wavenumber, temporal frequency, and convective velocity. This variability strongly impacts the estimated radiation pattern according to the model reported in Eq. 3. The contribution to the emission is not directly linked to the kinetic energy content of the HPOD modes used in the estimation, as reported in the literature, but it is strongly influenced by the actual parameters associated with the sound source contribution. In particular, according to the model used, the maximum levels of emission would be determined by the lengthscale of the respective wavepacket. Parameters as the hydrodynamic wavenumber and convective Mach number, instead, would be affecting mostly the emission pattern along the zenithal

direction.

It is worth remarking that the proposed technique can be applied also in the absence of time-resolved data to extract the spatial patterns of the wavepacket sources. This would still provide access to the spatial features of the comprised patterns (i.e. lengthscale and wavenumber), thus providing some insights into the contributions which might contribute mostly to the noise emission. Temporal resolution is indeed required to provide information on the temporal frequency and convective velocity of the wave. While this information would be lost in most of the cases from low-repetition-rate PIV, it is possible to retrieve it by pairing PIV with high-repetition-rate sensors (either pressure sensors or microphones, for example) using Extended-POD-based [16] (or more advanced [17]) methodologies for dynamic estimation of flow fields.

Additionally, an envisioned application of the space-only HPOD methodology is in conjunction with microphone array beamforming. Decomposing the beamforming map can provide better understanding of the noise sources. An example is provided in [22], where beamforming map POD is used to study interaction noise in fans. Leveraging the sound source patterns obtained through space-only HPOD on velocimetry data it could be possible to identify corresponding patterns in the beamforming maps: this could help bridge the gap between flow field structures and sound emission, providing additional tools for the study of jet noise.

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